

Distilling an LLM Teacher into a Compact Transformer for Toxic Comment Classification

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Abstract

This project studies knowledge distillation for Natural Language Processing through an “apprentice model”. The goal is to investigate if a compact student model can imitate the decision behaviour of a larger teacher model on a specific text classification task. The selected task is toxic comment classification. The experimental pipeline compares a classical TF-IDF and Logistic Regression baseline, a supervised BERT tiny student model, that was trained on original dataset labels, and an LLM OpenAI teacher used to relabel a controlled subset of examples. The supervised BERT-tiny student improves over the TF-IDF baseline in accuracy and F1 score, but is larger and slower at inference time. The final distilled student achieves an F1 score of 0.6962 against teacher labels. It shows partial imitation of the teacher’s decision boundary. The OpenAI teacher is much slower to query in the measured labelling pipeline, while the distilled student provides a compact local approximation for the selected toxicity classification task.

Keywords: Natural Language Processing, Knowledge Distillation, Toxicity Detection, BERT, TF-IDF, Large Language Models, GPT

Generative AI disclaimer

Generative AI, specifically GPT-5.5, was used during the project implementation to validate the overall approach, suggest ideas for improving the experimental results, check the grammar of the report, create LaTeX tables, generate experiment configuration files, and help resolve Python package conflicts in the virtual environment.

1 Introduction

Large language models have demonstrated strong performance across many Natural Language Processing tasks, including classification, question answering, summarisation, and reasoning. However, their size and computational cost make them difficult to utilize in constrained environments, especially when the target application requires limited resources. Knowledge distillation resolves this problem by transferring knowledge from a larger teacher model to a smaller student model [1]. Instead of trying to reproduce the full generality of the teacher, the student can be trained to imitate the teacher’s behaviour on a specific task or domain.

This project explores this idea in the context of toxic comment classification. Toxicity detection is a relevant NLP task because it combines text classification, contextual language understanding, and ethical considerations related to bias, moderation, and disagreement. The task is also appropriate for studying the tradeoff between model performance and efficiency. A large LLM may provide high quality judgments but it’s slow and expensive to use. Meanwhile, a compact model can be executed locally and quickly.

The project investigates if a compact Transformer model can be used as a lightweight student of a larger teacher model. The selected student is BERT-tiny, a small Transformer encoder. The teacher labelling stage uses an OpenAI LLM to classify examples and provide confidence scores. In addition, a classical TF-IDF and Logistic Regression model is used as a baseline, connecting the project to traditional vector space methods covered in the course.

The relevant literature includes the original formulation of knowledge distillation by Hinton et al. [1], where a student model learns from softened teacher output distributions and Transformer based language models such as BERT [2]. TinyBERT [3] demonstrate that smaller Transformer models can preserve part of the performance of larger models, reducing computational cost. This project applies the same general motivation to a practical toxicity detection pipeline and evaluates the resulting trade offs in task performance, model size, and inference speed.

2 Research question and methodology

2.1 Research question

The main research question is:

To what extent can a compact Transformer model imitate a larger LLM teacher for toxic comment classification. What are the trade offs between predictive performance, model size, and inference speed?

This question is task specific. The goal is not to show that a small model can replace a general purpose LLM in reasoning or conversation. The project studies if the teacher’s decision behaviour can be compressed into a smaller model for one domain specific classification problem.

2.2 Problem definition

Let x_i be a text comment and let $y_i \in \{0, 1\}$ be its original dataset label, where 0 denotes not toxic and 1 denotes toxic. The supervised baseline learns a classifier

$$f_\theta(x_i) \rightarrow \hat{y}_i, \quad (1)$$

using the original dataset labels.

In the teacher labelling stage, a large language model teacher T assigns a new label $\tilde{y}_i \in \{0, 1\}$ to each selected example of the data and confidence scores $p_T(y = 0|x_i)$ and $p_T(y = 1|x_i)$. The resulting teacher labelled dataset is

$$\mathcal{D}_T = \{(x_i, \tilde{y}_i)\}_{i=1}^N. \quad (2)$$

The distilled student S_ϕ is trained to imitate the teacher labels. In this implementation, I use hard-label distillation, where the loss is standard cross-entropy computed against the teacher label:

$$\mathcal{L}_{distill} = \mathcal{L}_{CE}(\tilde{y}_i, p_S(x_i)). \quad (3)$$

This differs from classical soft label distillation, where the student is trained to match the full teacher probability distribution. It can be implemented using a temperature scaled Kullback–Leibler divergence loss. A common formulation is

$$\mathcal{L} = (1 - \alpha) \mathcal{L}_{CE}(y, p_S) + \alpha \tau^2 \mathcal{L}_{KL}(p_T^{(\tau)}, p_S^{(\tau)}), \quad (4)$$

where $\mathcal{L}_{CE}$ is cross entropy with hard labels, $\mathcal{L}_{KL}$ is the Kullback–Leibler divergence between teacher and student probability distributions, τ is the distillation temperature, and α controls the trade off between the two terms. Soft label distillation is considered as future work in this project. In this project the implemented experiment focuses on hard label teacher imitation.

2.3 Pipeline overview

The project pipeline consists of four stages:

1. **Data preparation.** The dataset is loaded from Hugging Face and converted into reproducible train, validation, and test CSV splits. Because the original dataset is quite imbalanced, controlled stratified subsets are created.
2. **Baseline models.** A TF-IDF and Logistic Regression model is trained as a baseline. A BERT-tiny model is then finetuned on the original labels as a supervised Transformer baseline.
3. **Teacher labelling.** An OpenAI LLM teacher labels a stratified subset of examples and returns a binary label, confidence scores, and a short reason. This step produces a teacher labelled dataset for distillation.
4. **Distilled student.** Train a distilled BERT-tiny student on teacher labels, initialized from the supervised BERT-tiny run. Evaluate quality metrics and benchmark model size and inference speed.

The implementation is organised as a Python package. Separate modules are used for data loading, preprocessing, baseline models, Transformer training, teacher prompting, evaluation, and benchmarking.

3 Experimental results

3.1 Dataset and preprocessing

The experiments use the `SetFit/toxic_conversations` dataset from Hugging Face. The dataset contains comments labelled as toxic or not toxic and the columns `text`, `label`, and `label_text`. The full dataset contains more than 1.7 million training examples and 50,000 test examples. The original distribution is highly imbalanced, with around 8% toxic examples. To avoid wrong accuracy scores and to make the experiments faster and more controlled, stratified subsets were created with a 70 to 30 ratio between not toxic and toxic examples. For the main original label experiments, the following processed splits were used: 5000 training examples, 1000 validation examples, and 1000 test examples. For OpenAI teacher labeling, final deterministic teacher label splits were created from already labeled examples. Their distributions are reported in Table 2.

Preprocessing was intentionally minimal. Handled empty values, leading and trailing spaces were removed, and repeated spaces were normalised. No lowercasing or punctuation removal was applied, because Transformer tokenizers are designed to process raw text. The teacher consistently assigns more examples to the toxic class than the original annotations. The original split is fixed at 70% not toxic and 30% toxic. But the teacher distribution is closer to 55% not toxic and 45% toxic. This shows that the teacher applies a stricter toxicity boundary, especially for hostile or aggressive comments originally labeled as not toxic.

3.2 Evaluation metrics

The main classification metrics are accuracy, precision, recall, F1 score, and the confusion matrix. Because the task is class imbalanced, F1 score is treated as more informative than accuracy measure. In addition, model compactness and computational cost are evaluated using model size on disk, average inference time per example, and examples processed per second on CPU.

3.3 Baselines and supervised student

The first baseline is a classical TF-IDF and Logistic Regression classifier. It represents the vector space and linear classification part of the NLP course. The second model is a supervised BERT-tiny student based on `prajjwal1/bert-tiny`. It's trained on the original dataset labels. Table 1 shows the test results on the 1000 example of the original label test split.

The supervised BERT-tiny student improves over the TF-IDF baseline in test accuracy and F1 score. The improvement is mostly due to higher precision. BERT-tiny produces fewer false positives than the TF-IDF model. However, its recall is slightly lower, which means that it is more conservative in predicting toxicity and misses

Table 1 Performance on the original label 1000 example test split.

Model	Accuracy	Precision	Recall	F1
TF-IDF + Logistic Regression	0.7780	0.6327	0.6200	0.6263
BERT-tiny supervised student	0.8120	0.7240	0.6033	0.6582

slightly more toxic examples. This result motivates to use the compact Transformer models. And the TF-IDF baseline is still much smaller and faster.

The test confusion matrix for the TF-IDF baseline is

$$\begin{bmatrix} 592 & 108 \\ 114 & 186 \end{bmatrix}, \quad (5)$$

while the BERT-tiny supervised student obtains

$$\begin{bmatrix} 631 & 69 \\ 119 & 181 \end{bmatrix}. \quad (6)$$

These matrices show that BERT-tiny reduces false positives from 108 to 69, but slightly increases false negatives from 114 to 119.

3.4 Teacher labelling analysis

An OpenAI LLM teacher was used to label controlled stratified subsets of the data. The teacher prompt defined toxicity as insults, harassment, threats, hate, severe disrespect, or abusive language. It also stated that comments may be negative, political, or disagreeable without necessarily being toxic. The teacher returned a binary label, confidence scores, and a short reason. On the final teacher label test split, agreement between original and teacher labels is 0.8033. Table 2 summarises the current teacher labelled subset.

Table 2 Agreement between original dataset labels and OpenAI teacher labels on the final teacher-labelled splits.

Split	Rows	Original 0	Original 1	Teacher 0	Teacher 1	Agreement
Train	1500	1050	450	816	684	0.7627
Validation	300	210	90	160	140	0.7733
Test	300	210	90	167	133	0.8033

Two main disagreement patterns were observed:

- Original label 0, teacher label 1: comments that were originally annotated as non-toxic but contained hostile, demeaning, insulting, or identity-related language.

- Original label 1, teacher label 0: comments that contained sensitive, political, or negative terms, but did not necessarily contain direct harassment or abuse under the prompt definition.

The teacher agrees with the original labels on 76.27% of the train split, 77.33% of the validation split, and 80.33% of the test split. The teacher labels a higher proportion of examples as toxic than the original dataset. This suggests that the teacher uses a stricter decision boundary for hostile language. For example, comments originally labelled as nontoxic but containing phrases such as “you do not have a clue”, “vile race baiting hate propaganda”, or “more yapping from someone who knows nothing” were often labelled as toxic by the teacher. Conversely, some comments originally labelled toxic were considered nontoxic by the teacher when they contained mild profanity or negative language without direct abuse.

This confirms that the LLM teacher should not be treated as a gold standard. Instead, it defines a different decision boundary. The distillation task is therefore best interpreted as teacher imitation, not a direct optimization of the original dataset labels.

3.5 Distilled student results

The first distillation attempt was trained from the base BERT-tiny model on a smaller teacher labeled set. It predicted too few toxic examples and performed poorly. The final approach improved this by using more teacher labeled examples and initializing the distilled model from the supervised BERT-tiny checkpoint. The resulting model was then finetuned on OpenAI teacher labels.

Table 3 reports the final distilled student results. The model is evaluated in comparison to two cases. First, against teacher labels that measures teacher imitation. Second, against original labels, that measures compatibility with the original dataset annotations.

Table 3 Final distilled BERT-tiny results on the teacher labeled validation and test subsets.

Split	Evaluation target	Accuracy	F1
Validation	Teacher labels	0.7433	0.7586
Test	Teacher labels	0.7033	0.6962
Test	Original labels	0.7067	0.6480

The distilled student reaches an F1 score of 0.6962 against the teacher on the final test subset. This indicates that the compact model partially reaches the teacher’s decision boundary. Against the original labels on the teacher-labelled test subset, its F1 score is 0.6480. This is close to the supervised BERT-tiny result of 0.6582 obtained on the larger 1000-example original-label test split. But the two numbers are not perfectly comparable because they are computed on different test subsets. This is expected because the distilled student is optimized for teacher imitation, and the

teacher disagrees with the original labels on approximately 20% of the final teacher labeled test subset.

The prediction distribution also shows a limitation. On the final test subset, the original labels contain 210 nontoxic and 90 toxic examples. The teacher labels contain 167 nontoxic and 133 toxic examples. The distilled model predicts 140 nontoxic and 160 toxic examples. Therefore, the student successfully moves toward the teacher’s stricter toxicity boundary. But it slightly overpredicts the toxic class compared with the teacher.

3.6 Efficiency benchmark

Efficiency was measured on CPU using the test split. The benchmark includes model size on disk, average inference time per example, and throughput in examples per second. Results are shown in Table 4.

Table 4 CPU efficiency benchmark for locally deployable models

Model	Size (MB)	Time per example (ms)	Examples per sec
TF-IDF and Logistic Regression	0.89	0.0611	16379.73
BERT-tiny supervised student	17.42	1.6061	622.61
BERT-tiny distilled student	17.42	1.5578	641.94

The TF-IDF baseline is approximately 19.6 times smaller than the BERT-tiny models and more than 25 times faster on CPU. However, its original label F1 score is lower than the supervised BERT-tiny model. The supervised and distilled BERT-tiny models have almost identical size and speed. It’s expected, because they share the same architecture. Their difference is the decision boundary learned during training. The supervised model follows the original labels, while the distilled model adapts to the teacher labels.

Table 5 Practical comparison between the OpenAI teacher and the distilled student.

Model	Deployment	Time per example	Examples per sec
OpenAI LLM teacher	Remote API	5440 ms	0.18
BERT-tiny distilled student	Local CPU	1.5578 ms	641.94

The OpenAI teacher is not directly comparable to the local models in terms of model size. Because it is accessed through a proprietary remote API and its internal parameters and serving hardware are not observable. However, the practical labelling time can be estimated from the teacher labelling runs. Labelling 2100 examples took approximately 3.17 hours, corresponding to about 5.44 seconds per example, or 0.18 examples per second. In contrast, the distilled BERT-tiny student runs locally on CPU

at 1.5578 ms per example, or 641.94 examples per second. Therefore, the distilled student is approximately 3500 times faster in the measured deployment setting.

This comparison illustrates the motivation for distillation. The OpenAI teacher is useful for producing high-level labels, confidence scores, and short explanations. But it is costly and slow to query repeatedly. The distilled student does not reproduce the full capabilities of the LLM, but it provides a compact local approximation of the teacher’s behaviour on the selected toxicity classification task.

4 Concluding Remarks

In this project a complete apprentice model pipeline was implemented and evaluated for toxic comment classification. The results show that a compact BERT-tiny student can be adapted to imitate an LLM teacher on a narrow NLP task. The final distilled model achieved an F1 score of 0.6962 against teacher labels, demonstrating partial transfer of the teacher’s decision behaviour. At the same time, the distilled model did not improve performance against the original dataset labels compared with the supervised BERT-tiny baseline. Because the teacher applies a different and stricter interpretation of toxicity, so optimizing for teacher imitation naturally changes the evaluation behaviour.

The experiments also highlight important tradeoffs. The TF-IDF baseline is extremely efficient and interpretable, but weaker in F1 score. BERT-tiny provides better classification quality, but is significantly larger and slower. The OpenAI teacher, on the other hand, is powerful and can provide labels, confidence scores, and short explanations. But it is accessed through a remote API and is expensive and slow to query repeatedly. In the measured labelling pipeline, the OpenAI teacher required approximately 5.44 seconds per example, while the distilled BERT-tiny student performed local CPU inference in 1.5578 ms per example. This illustrates the practical motivation for distillation: expensive teacher decisions can be used offline to train a compact local model that is much faster at deployment time. Distillation does not change model size or speed because the architecture remains the same, but it changes the decision boundary of the student. This supports the idea that knowledge distillation can be used not only for compression, but also for task specific behavioural adaptation.

There are several limitations. First, the project uses hard label distillation, not aclassical soft label distillation with temperature and KL-divergence. This loses information contained in the teacher’s probability distribution. Second, the OpenAI teacher is not an objective ground truth and may be sensitive to prompt wording. Third, the final teacher labeled dataset is smaller than the original supervised training set because API labeling has cost and time constraints. Finally, the distilled student slightly over-predicts the toxic class compared with the teacher. It shows that threshold tuning or soft label training could improve calibration.

Future work could extend the project in several directions. The most direct extension is soft label distillation using teacher probabilities and a KL-divergence loss. Another extension is threshold calibration on the validation set to better match the teacher’s label distribution. Additional work could compare multiple teachers, such as an OpenAI LLM and an open-source toxicity classifier. Or include explanation

based distillation. Finally, a larger teacher labelled dataset would allow a more reliable evaluation of generalization and teacher imitation.

References

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